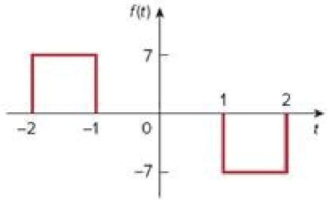
Chapter 18, Problem 1P

Problem

Obtain the Fourier transform of the function in Fig. 18.26.

Figure 18.26



Step-by-step solution

Step 1 of 2

Refer to the function $f(t)$ in Figure P18.26 in the textbook.

The function $f(t)$ is expressed as,

$$f(t)=\begin{cases}7, & -2\le t\le -1\\ 0, & -1<t<1\\ -7, & 1\le t\le 2\end{cases}$$

The Fourier transform of the function $f(t)$ is,

$$\begin{aligned}F(\omega)&=\int_{-\infty}^{\infty} f(t)e^{-j\omega t}dt\\ &=\int_{-2}^{-1}7e^{-j\omega t}dt+\int_{-1}^{1}(0)e^{-j\omega t}dt+\int_{1}^{2}(-7)e^{-j\omega t}dt\\ &=\int_{-2}^{-1}7e^{-j\omega t}dt-\int_{1}^{2}7e^{-j\omega t}dt\\ &=7\left[\frac{e^{-j\omega t}}{-j\omega}\right]_{-2}^{-1}-7\left[\frac{e^{-j\omega t}}{-j\omega}\right]_{1}^{2}\end{aligned}$$

Step 2 of 2

Further simplify.

$$
\begin{aligned}
F(\omega) &= -\frac{7}{j\omega}\left[e^{j\omega} - e^{j2\omega}\right] + \frac{7}{j\omega}\left[e^{-j2\omega} - e^{-j\omega}\right] \\
&= -\frac{7}{j\omega}\left[e^{j\omega} - e^{j2\omega}\right] + \frac{7}{j\omega}\left[e^{-j2\omega} - e^{-j\omega}\right] \\
&= \frac{7}{j\omega}\left[e^{-j2\omega} + e^{j2\omega} - e^{-j\omega} - e^{j\omega}\right] \\
&= \frac{7}{j\omega}\left[e^{j2\omega} + e^{-j2\omega} - \left(e^{j\omega} + e^{-j\omega}\right)\right]
\end{aligned}
$$

Since $\cos\omega = \dfrac{e^{j\omega} + e^{-j\omega}}{2}$,

$$
\begin{aligned}
F(\omega) &= \frac{7}{j\omega}\left[2\cos 2\omega - 2\cos\omega\right] \\
&= \frac{14(\cos 2\omega - \cos\omega)}{j\omega}
\end{aligned}
$$

Thus, the Fourier transform of the function $f(t)$ is $\boxed{\dfrac{14(\cos 2\omega - \cos\omega)}{j\omega}}$.